

Physics-Informed PMU Event Detection

and Localization

How the submitted pipeline turns raw PMU data into
event and location predictions

| sgms_extra_trees_windowed_v2 | | 30 s windows | | ExtraTrees |

Final variant: base_v2 + rolling + rls_kalman + graph_temporal



SGSMA 2026 Challenge

Main story

1. Convert raw PMU waveforms into physics-aware features.
2. Use a hierarchical ExtraTrees classifier to identify event type.
3. Route each event to a specialized BUS, LINE, or PMU localizer.
4. Produce the challenge CSV output at timestamp resolution.

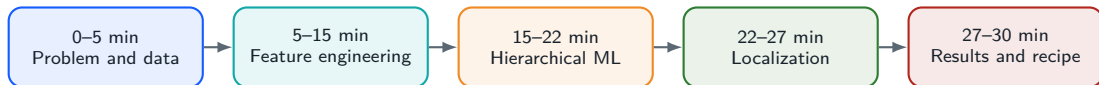
Hands-on goal

After this talk, you should be able to reimplement the method from scratch if a new dataset is released during the on-site round.

Submission boundary

The final submitted method was SGMS, not the leaky DeepSeek RAW-tuned variants.

30-minute presentation map



Backup slides

Detailed thresholds, artifacts, event-specific recipes, validation commands, and generalization notes are included after the main deck.

Section 1

Problem and Data



Input

Multiple PMU CSV files, usually one per observed bus:

- ▶ synchronized timestamps,
- ▶ voltage and current phasors,
- ▶ frequency and ROCOF,
- ▶ data-quality indicators.

Output

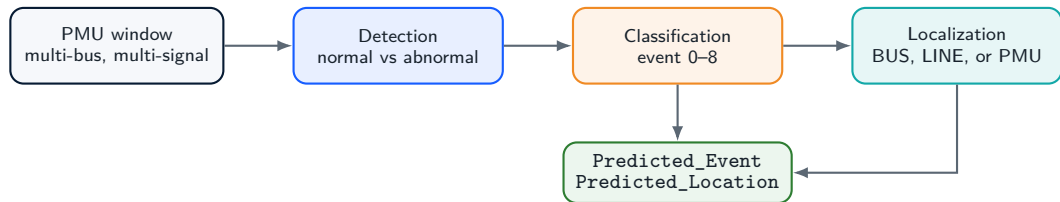
The submission must generate exactly:

- ▶ `TIMESTAMP`
- ▶ `Bus`
- ▶ `Predicted_Event`
- ▶ `Predicted_Location`

Scoring target

Detection is not enough. The system must also classify the event type and localize its origin or the failed PMU.

The task is three coupled decisions



Design consequence

The pipeline keeps these decisions separate. That is why the final model has a physical classifier, missing/bad-data detectors, and typed localizers instead of one flat classifier.

Label	Name	Physical interpretation
0	normal	No physical disturbance and no major data issue.
1	fault	Strong voltage disturbance, fast derivative, sag or transient.
2	line outage	Current redistribution without fault-level voltage collapse.
3	generation change	Frequency/ROCOF and active-power proxy change.
4	load change	Sustained current and power-consumption change.
5	missing data	Data loss without strong physical evidence.
6	missing + physical	Data loss combined with a real physical event.
7	bad data	One sensor/channel becomes inconsistent with the rest.
8	mixed proxy	Mixed physical signature not cleanly explained by 1–4.

Electrical measurements

Group	Columns
Voltage magnitude	VA_MAG, VB_MAG, VC_MAG
Current magnitude	IA_MAG, IB_MAG, IC_MAG
Voltage angle	VA_ANG, VB_ANG, VC_ANG
Current angle	IA_ANG, IB_ANG, IC_ANG
Dynamics	Freq, ROCOF

Support variables

- ▶ **TIMESTAMP**: time alignment.
- ▶ **DATA_PRESENT**: measurement availability.
- ▶ Bus ID from `Bus*.csv`.
- ▶ IEEE 39-bus topology for graph features.

Interpretation

Every feature family is built from a physical question: how large, how fast, how persistent, how coherent, and where in the grid?

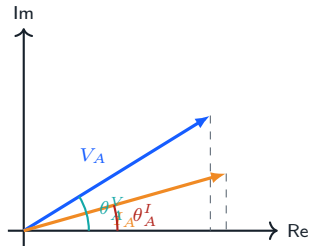
Per bus, per phase

$$V_{\phi}(t) = |V_{\phi}(t)|e^{j\theta_{\phi}^V(t)}, \quad I_{\phi}(t) = |I_{\phi}(t)|e^{j\theta_{\phi}^I(t)}$$

$$\phi \in \{A, B, C\}, \quad \dot{f}(t) = \text{ROCOF}(t)$$

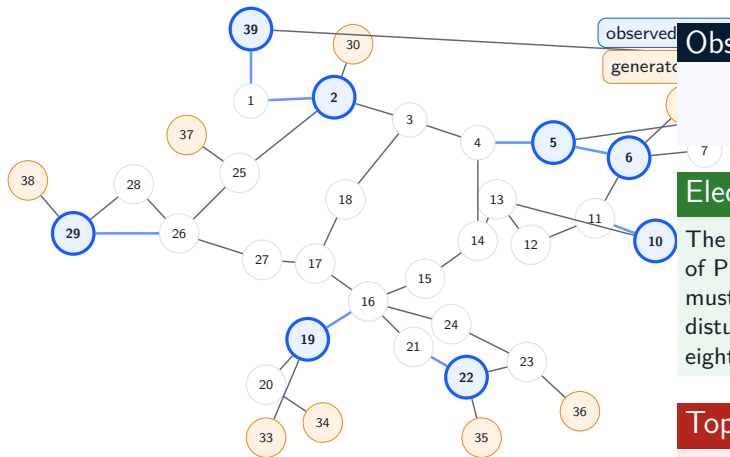
What we feed to features

The extractor uses magnitudes, unwrapped angles, frequency, ROCOF, and data availability. It does not solve a full AC power-flow at inference time.



Power proxy comes from the angle gap $\theta^V - \theta^I$ and normalized magnitudes.

IEEE 39-bus system view



Observed PMUs

$\{2, 5, 6, 10, 19, 22, 29, 39\}$

Electrical reading

The localizer sees only a sparse set of PMU views. A line or bus event must be inferred from how the disturbance appears across those eight locations.

Topology used

The repo builds electrical distances from the IEEE 39-bus admittance model, not from Euclidean drawing

Admittance and impedance

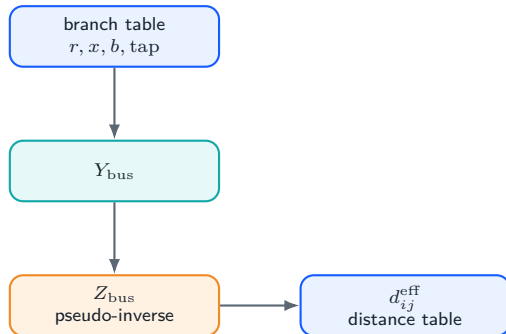
Y_{bus} : network admittance matrix

$$Z_{\text{bus}} = Y_{\text{bus}}^+$$

$$d_{ij}^{\text{eff}} = |Z_{ii} + Z_{jj} - Z_{ij} - Z_{ji}|$$

Meaning

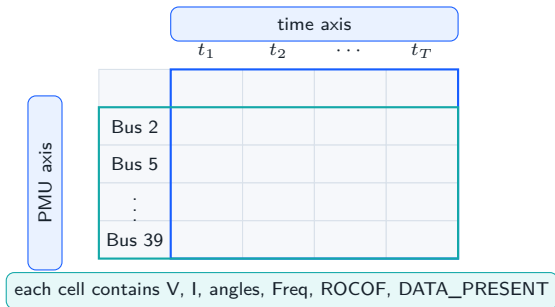
Two buses are close if an injection at one bus is expected to have a similar voltage effect at the other.



Where it appears

Graph-temporal features weight PMU differences by $1/d_{ij}^{\text{eff}}$.

How a case is represented before feature extraction



The extractor reads a tensor

Think of one case as a table over PMUs and time. Each table cell has several electrical channels.

What changes after extraction

The window becomes one row of engineered features. Models never see the raw tensor directly.

Observed PMUs in the final configuration



Bus 2

Bus 5

Bus 6

Bus 10

Bus 19

Bus 22

Bus 29

Bus 39

Why PMU placement matters

Localization relies on spatial fingerprints. A disturbance near a bus or line affects nearby PMUs more strongly and often earlier than remote PMUs.

Generalization note

The runtime discovers PMU files from `Bus*.csv`; it does not hardcode RAW0001 placement in the submitted path.

Entrypoint

```
python main.py -input-dir <folder> -output  
<csv>
```

Active route

- ▶ `model = sgms_extra_trees_windowed_v2`
- ▶ `route = ml_windowed`
- ▶ `window_seconds = 30`

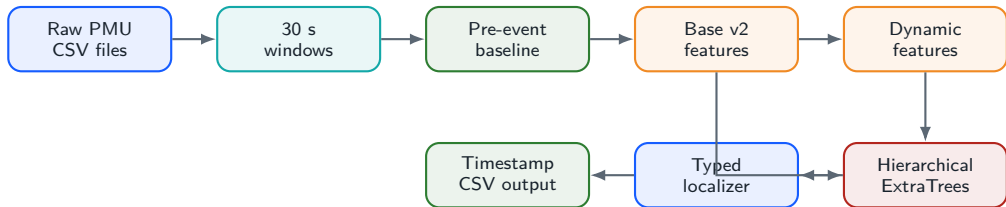
Call chain

```
main.py  
→ src/models/hybrid_submission.py  
→ src/data_factory/final_model.py
```

Fallback

A bus-agnostic physics runtime remains only as a missing-bundle fallback.

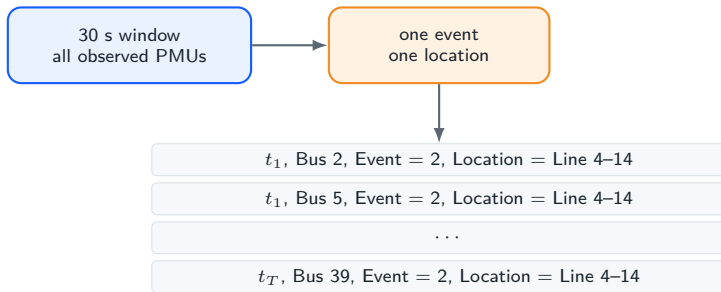
End-to-end method at a glance



Runtime shape

We convert each PMU window into a physics-aware signature, classify the event with guarded ExtraTrees models, then localize it with event-specific localizers.

Window prediction expands back to timestamp rows



Submission contract

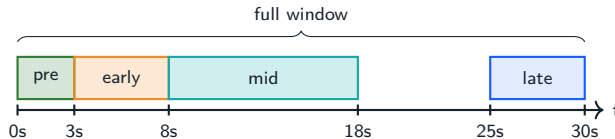
The official CSV is timestamp-aligned even though the decision is made at the window level.

Section 2

Windowing and Baseline



Why a 30-second window?



Reasoning

- ▶ One sample is too noisy.
- ▶ A fault has onset and transient.
- ▶ Load/generation changes are often slower.
- ▶ Missing data needs run-length evidence.

Implementation

The model predicts once per window and replicates the window prediction to all required timestamp/bus rows.

Segment	Time span	Why it exists
pre	first 3 seconds	Estimate normal operating point.
early	3–8 seconds	Capture onset and fast transient.
mid	8–18 seconds	Capture stabilization and oscillation.
late	last 5 seconds	Capture persistent post-event level.
full	full 30 seconds	Capture total severity and duration.

Window reading

The same signal can mean different things depending on when it changes. Timing is part of the physics.

Given one 30-second window

$$t_0 = \min(t), \quad t_1 = \max(t), \quad t_p = t_0 + 3$$

$$\mathcal{T}_{pre} = \{t : t \leq t_p\}$$

$$\mathcal{T}_{early} = \{t : t_p < t \leq t_p + 5\}$$

$$\mathcal{T}_{mid} = \{t : t_p + 5 < t \leq t_p + 15\}$$

$$\mathcal{T}_{late} = \{t : t \geq \max(t_p, t_1 - 5)\}$$



Why this matters

Event labels are window labels, but electrical signatures are time-shaped. These masks let us compare the same signal before and after the disturbance.

Baseline per PMU and signal

$$b_x = \text{median}(x(t), t \in \text{pre})$$

Magnitude normalization

$$x_{\text{pu}}(t) = \frac{x(t)}{b_x + \epsilon}$$
$$\Delta x(t) = x(t) - b_x$$

Angle handling

Angles are unwrapped before computing differences:

$$\theta_{\text{clean}} = \text{unwrap}(\theta)$$

This avoids artificial jumps near the angle boundary.

Baseline and robust scale are computed locally



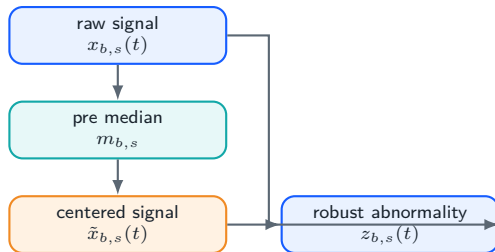
For each bus b and signal s

$$m_{b,s} = \text{median}_{t \in \mathcal{T}_{pre}} x_{b,s}(t)$$

$$\tilde{x}_{b,s}(t) = x_{b,s}(t) - m_{b,s}$$

$$\sigma_{b,s}^{rob} = 1.4826 \cdot \text{median}_{t \in \mathcal{T}_{pre}} |x_{b,s}(t) - m_{b,s}|$$

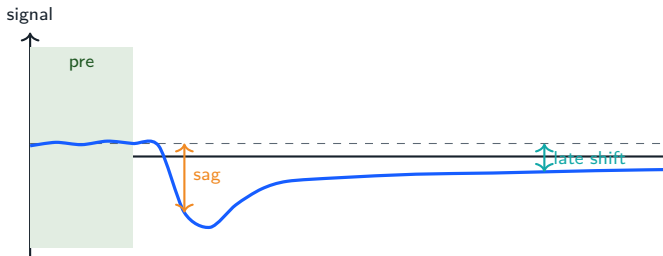
$$z_{b,s}(t) = \frac{|x_{b,s}(t) - m_{b,s}|}{\max(\sigma_{b,s}^{rob}, \epsilon)}$$



Implementation detail

If the MAD is too small, the code falls back to a standard deviation scale. If that also fails, it uses a safe fallback value.

Baseline subtraction, visually



What the model receives

The feature extractor does not keep this curve. It records quantities such as early span, derivative, maximum deviation, and late shift.

Fault reading

A sharp voltage sag after the pre segment creates large derivative and large robust z-score features.

Canonical pattern

`<PMU/BUS>__<SIGNAL>__<SEGMENT>__<STATISTIC>`

Examples

`Bus2__VA_MAG__early__span`
`Bus10__Freq__late__mean`
`GLOBAL__single_signal_score_max`

Reimplementation note

The feature vector has tens of thousands of columns. A consistent naming scheme makes debugging and reimplementation possible.

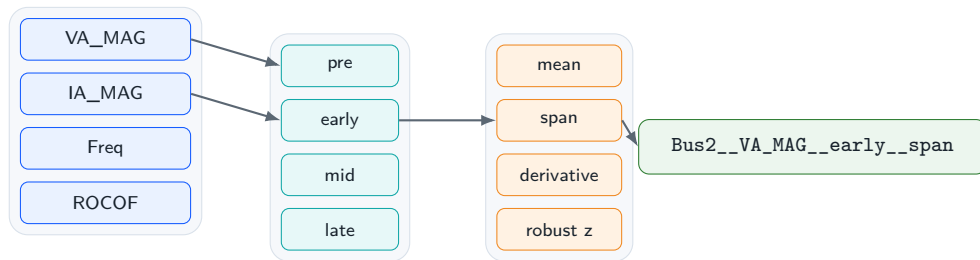
Section 3

Base Feature Engineering



Family	Computed from	Physical question
Segment statistics	every signal and segment	How large and variable was the response?
Temporal deltas	early/mid/late vs pre	Did the operating point move?
Derivatives	finite differences	How abrupt was the event?
Robust z-score	pre-event median/MAD	How abnormal was the signal?
Data quality	gaps, NaNs, data-present	Is this a measurement failure?
Single-signal anomaly	ranked abnormal channels	Is one channel corrupted?
Power proxy	V/I magnitudes and angles	Did active/reactive power change?
Global aggregations	PMU-level feature summaries	What is the system-wide pattern?

Feature factory: signal $\times$ segment $\times$ statistic



Reimplementation shortcut

Write the extractor as nested loops: for each PMU, for each signal, for each segment, compute the same statistics with the same names.

For centered values $\tilde{x}(t)$ in segment $\mathcal{S}$

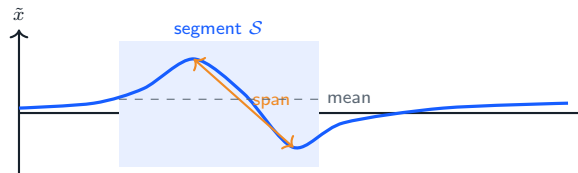
$$\mu_{\mathcal{S}} = \frac{1}{|\mathcal{S}|} \sum_{t \in \mathcal{S}} \tilde{x}(t)$$

$$\sigma_{\mathcal{S}} = \sqrt{\frac{1}{|\mathcal{S}|} \sum_{t \in \mathcal{S}} (\tilde{x}(t) - \mu_{\mathcal{S}})^2}$$

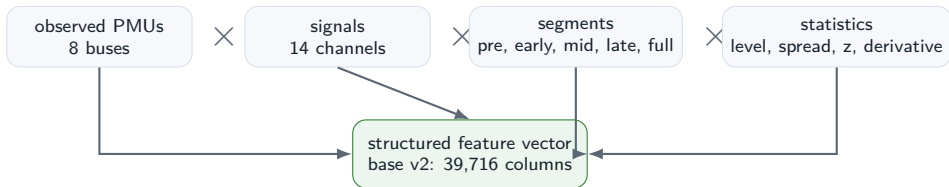
$$\text{span}_{\mathcal{S}} = \max_{t \in \mathcal{S}} \tilde{x}(t) - \min_{t \in \mathcal{S}} \tilde{x}(t)$$

Also stored

min, max, median, $\max |\tilde{x}|$, $\text{mean}|\tilde{x}|$, $p05$, $p95$



Feature growth is controlled by loops



Programming recipe

Do not hand-write features. Generate names and values from loops, then freeze the resulting column order for training and inference.

1. Segment statistics



For each signal and each segment

{mean, std, min, max, span, median, max_abs, mean_abs, p05, p95}

Physical meaning

- ▶ **span**: event severity.
- ▶ **std**: oscillation or instability.
- ▶ **p05/p95**: stable extremes.
- ▶ **mean/median**: level shift.

Event use

Faults often create high voltage span; line outages produce current-span changes; generation/load events often produce sustained mean shifts.

2. Temporal deltas



Definitions

$$\Delta_{\text{early}} = \mu_{\text{early}} - \mu_{\text{pre}}$$

$$\Delta_{\text{mid}} = \mu_{\text{mid}} - \mu_{\text{pre}}$$

$$\Delta_{\text{late}} = \mu_{\text{late}} - \mu_{\text{pre}}$$

Interpretation

- ▶ early delta: onset shock.
- ▶ mid delta: transient settling.
- ▶ late delta: persistent operating-point change.

Why not only full-window stats?

Two windows can have the same full-window mean but different temporal behavior. The deltas preserve event timing.

3. Derivative features



Finite-difference approximation

$$d_x(t) = \frac{x(t) - x(t - \Delta t)}{\Delta t}$$

$$\text{max_abs_derivative} = \max_t |d_x(t)|, \quad \text{rms_derivative} = \sqrt{\frac{1}{T} \sum_t d_x(t)^2}$$

High derivative

Usually means an abrupt transient: fault onset, line switching, or corrupted data.

Event use

The final rule engine uses voltage and current derivative thresholds to guard fault and line-outage decisions.

4. Robust z-score



Robust scale from the pre-event segment

$$m = \text{median}(x_{\text{pre}}), \quad \text{MAD} = \text{median}(|x_{\text{pre}} - m|)$$
$$z_{\text{robust}}(t) = \frac{|x(t) - m|}{\text{MAD} + \epsilon}$$

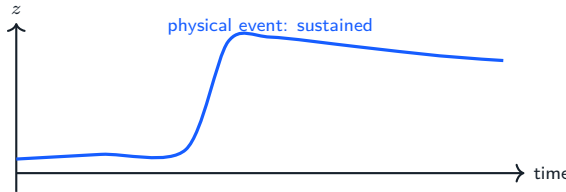
Features

- ▶ max_abs_robust_z
- ▶ sustained_abs_robust_z_5
- ▶ sustained_abs_robust_z_15

Why median/MAD here

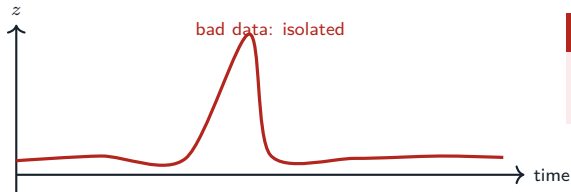
The median and MAD are less sensitive to spikes than mean and standard deviation, which matters when bad data is one of the event classes.

Robust z-score separates two failure modes



Why we keep both max and sustained scores

`max_abs_robust_z` catches the spike. Sustained features say whether the abnormal value stays high for several samples.



Classifier use

The pair helps the event model avoid calling every spike a physical event.

5. Sustained anomaly features



Concept

Not every high z-score is equally meaningful. A single spike may be bad data; a sustained high z-score may be a physical event.

Short sustained score

`sustained_abs_robust_z_5`

Captures short bursts and sharp events.

Long sustained score

`sustained_abs_robust_z_15`

Captures persistent deviations after the event.

Practical reading

Faults can be sharp and sustained; bad data is often isolated to one channel; load/generation changes persist in late features.

6. Data-quality features



Features

- ▶ `n_samples, duration_s, dt_s`
- ▶ `max_timestamp_gap_s`
- ▶ `max_timestamp_gap_ratio`
- ▶ `data_present_fraction`
- ▶ `data_present_min`
- ▶ `missing_run_max_samples`
- ▶ `missing_run_max_s`
- ▶ `nan_fraction_max/mean`

Decision logic

missing $\Rightarrow \begin{cases} \text{Event 5,} & \text{no physical event} \\ \text{Event 6,} & \text{physical event also present} \end{cases}$

Physical meaning

Missing-data events are communication or measurement failures, not necessarily grid disturbances.

7. Single-signal bad-data features



Goal

Detect a channel that becomes extreme while the rest of the PMU/grid remains coherent.

Features

- ▶ `single_signal_score_max`
- ▶ `single_signal_score_second`
- ▶ `single_signal_score_ratio`
- ▶ `single_signal_score_count_gt20`

Rule of thumb

$$\frac{\text{top anomaly}}{\text{second anomaly}} \geq 3 \Rightarrow \text{bad data candidate}$$

8. Phase-spread and consistency features



What is measured

For voltage/current magnitude and angle groups, the extractor summarizes phase-to-phase spread and consistency.

Physical role

- ▶ three-phase imbalance,
- ▶ phase-angle inconsistency,
- ▶ PMU channel mismatch.

Event use

These features help separate grid events from measurement artifacts because real events tend to preserve more cross-phase structure than corrupted channels.

Power proxy from voltage-current phase angle



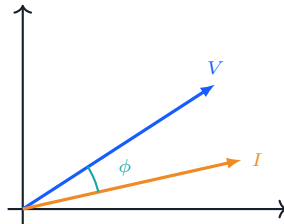
Per-unit magnitudes

$$V_{pu,\phi}(t) = \frac{|V_\phi(t)|}{\max(\text{median}_{pre}|V_\phi|, \epsilon)}$$

$$I_{pu,\phi}(t) = \frac{|I_\phi(t)|}{\max(\text{median}_{pre}|I_\phi|, \epsilon)}$$

Proxy, not calibrated power

The challenge data does not require a full power-flow solve. The proxy gives the classifier a stable signature of active/reactive behavior.



$$\begin{aligned} P_{\text{proxy}} &= V_{pu} I_{pu} \cos \phi \\ Q_{\text{proxy}} &= V_{pu} I_{pu} \sin \phi \end{aligned}$$

Electrical reading

A load or generation change can shift I , ϕ , and frequency while voltage magnitude remains moderate.

9. Power proxy features

Per phase

$$\phi = \theta_V - \theta_I$$

$$P_{\text{proxy}} = V_{\text{pu}} I_{\text{pu}} \cos(\phi)$$

$$Q_{\text{proxy}} = V_{\text{pu}} I_{\text{pu}} \sin(\phi)$$

$$PF_{\text{ang}} = \phi$$

Use in classifier

Generation and load events may not create dramatic voltage disturbances. They are often easier to see through frequency, current, and active/reactive power proxy changes.

Used most by

Event 3, Event 4, and mixed physical Event 8.

10. Global PMU aggregation



What is aggregated

After extracting bus-level features, the model also computes global summaries across observed PMUs:

mean, max, min, std

Why it helps

A local event has a spatial footprint. Global aggregation gives the classifier a system-level view: is the disturbance concentrated, widespread, coherent, or isolated?

Example

Bad data often has one extreme PMU/channel. A physical event affects multiple PMUs in a structured way.

Base feature count



Feature group	Count
Voltage angle	7,596
Voltage magnitude	7,596
Current angle	7,596
Current magnitude	7,596
Frequency / ROCOF	4,640
Power proxy	4,428
Data quality	156
Single-signal anomaly	48
Other/global	60
Base v2 total	39,716

Reimplementation note

The base model uses a large but structured feature vector. The structure is what makes it reimplementable.

Section 4

Dynamic Feature Engineering



Why dynamic features were added



Base features answer

How large, abnormal, and persistent was the signal within each segment?

Dynamic features answer

How did the signal evolve across time scales, and how did the disturbance propagate across PMUs?

Rolling

Multi-scale local behavior.

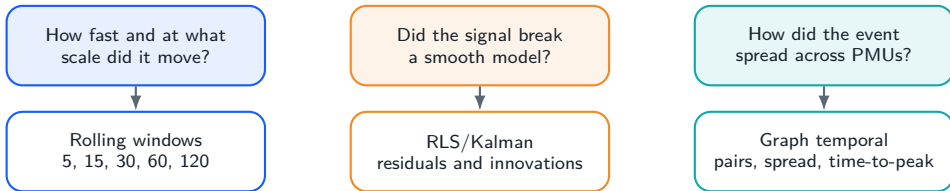
RLS/Kalman

Prediction residuals and innovations.

Graph temporal

Spatial-temporal PMU signature.

Dynamic blocks as engineering questions



Implementation reading

Each dynamic block is a small, testable module. You can add them one at a time after the base extractor works.

Main signal set

$\{VA/VB/VC_MAG, IA/IB/IC_MAG, VA/VB/VC_ANG, IA/IB/IC_ANG, Freq, ROCOF\}$

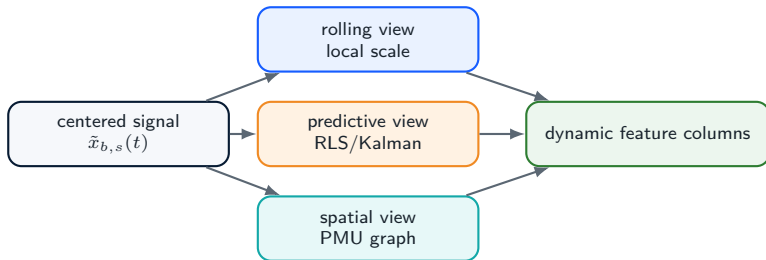
Reduced signal set for RLS/Kalman and graph features

$\{VA_MAG, IA_MAG, VA_ANG, IA_ANG, Freq, ROCOF\}$

Reason

The reduced set keeps dynamic features feasible while preserving representative voltage, current, angle, and frequency behavior.

Dynamic extractor: same signal, three views



Implementation reading

All dynamic blocks start after baseline subtraction. That makes the blocks comparable across buses and across RAW/SIM cases.

Rolling windows

$$w \in \{5, 15, 30, 60, 120\}$$

Features

- ▶ mean_abs_max
- ▶ std_max, std_mean
- ▶ slope_max_abs
- ▶ energy
- ▶ time_to_abs_peak

Physical interpretation

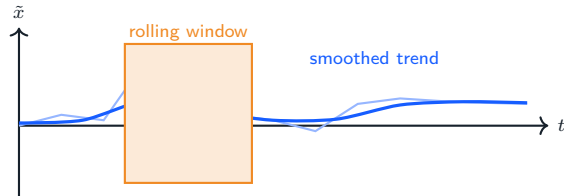
- ▶ short windows catch abrupt switching/fault onset;
- ▶ long windows catch sustained load/generation shifts;
- ▶ time-to-peak helps localize event propagation.

Centered rolling mean and standard deviation

$$\mu_w(t) = \frac{1}{w} \sum_{\tau \in W_w(t)} \tilde{x}(\tau)$$

$$\sigma_w(t) = \sqrt{\frac{1}{w} \sum_{\tau \in W_w(t)} (\tilde{x}(\tau) - \mu_w(t))^2}$$

$$E_x = \frac{1}{T} \sum_t |\tilde{x}(t)|^2$$



Stored for each w

$$\max |\mu_w|, \quad \max \sigma_w, \quad \overline{\sigma_w}, \quad \max \left| \frac{d\mu_w}{dt} \right|, \quad E_x, \quad t_{\text{peak}}$$

Concept

Fit a simple expected trajectory. If the actual signal deviates sharply from that expected trajectory, the residual or innovation becomes large.

RLS-style features

- ▶ `level_resid`
- ▶ `theta_delta`
- ▶ `linear_resid`
- ▶ forgetting factors: 0.92 and 0.985

Kalman-style features

- ▶ `kalman_innov`
- ▶ max, mean, p95
- ▶ time-to-peak

Model

$$y_t = \theta_t + e_t$$

$$K_t = \frac{P_{t-1}}{\lambda + P_{t-1}}$$

$$\theta_t = \theta_{t-1} + K_t(y_t - \theta_{t-1})$$

$$P_t = \frac{P_{t-1} - K_t P_{t-1}}{\lambda}$$

Forgetting factors

$$\lambda \in \{0.92, 0.985\}$$

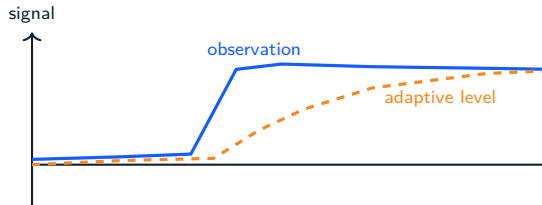
Features produced

$$r_t = |y_t - \theta_{t-1}|$$

$$\Delta\theta_t = |\theta_t - \theta_{t-1}|$$

For each sequence:

max, mean, p95, median_{late} - median_{pre}



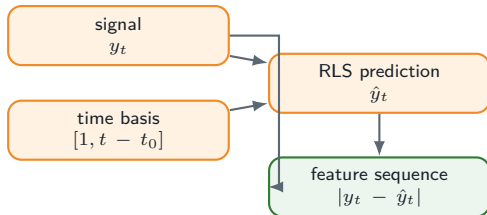
Local linear predictor

$$\phi_t = \begin{bmatrix} 1 \\ t - t_0 \end{bmatrix}, \quad \hat{y}_t = \phi_t^\top \theta_{t-1}$$

$$K_t = \frac{P_{t-1} \phi_t}{\lambda + \phi_t^\top P_{t-1} \phi_t}$$

$$\theta_t = \theta_{t-1} + K_t (y_t - \hat{y}_t)$$

$$P_t = \frac{P_{t-1} - K_t \phi_t^\top P_{t-1}}{\lambda}$$



Why linear residuals help

They discount smooth ramps but react to switching, faults, and sudden sensor errors.

State-space view

$$x_t = x_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, Q)$$

$$y_t = x_t + v_t, \quad v_t \sim \mathcal{N}(0, R)$$

$$Q = (0.005 \text{ s})^2, \quad R = (0.08 \text{ s})^2$$

$$s = \text{robust_scale}(y)$$

Update equations

$$P_t^- = P_{t-1} + Q$$

$$\nu_t = y_t - \hat{x}_{t-1}$$

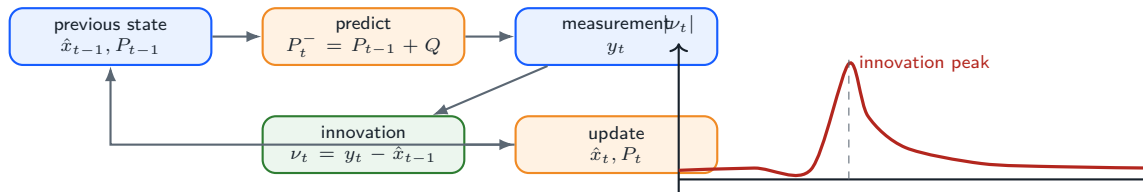
$$K_t = \frac{P_t^-}{P_t^- + R}$$

$$\hat{x}_t = \hat{x}_{t-1} + K_t \nu_t, \quad P_t = (1 - K_t) P_t^-$$

Feature sequence

The stored signal is $|\nu_t|$, the absolute innovation. A large innovation means the measurement did not match the smooth level model.

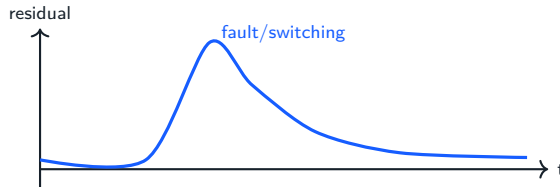
Kalman innovation diagram



Stored features

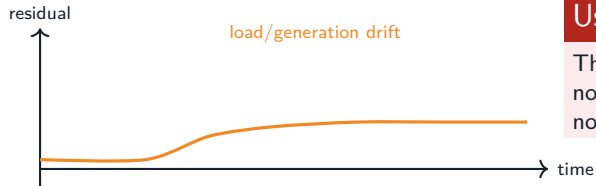
$\max |\nu_t|, \quad \text{mean} |\nu_t|, \quad p95 |\nu_t|, \quad t_{\text{peak}}$

How RLS/Kalman separates event shapes



What the model sees

Faults and switching tend to create a sharp innovation peak. Load and generation changes can create smaller but persistent residual deltas.

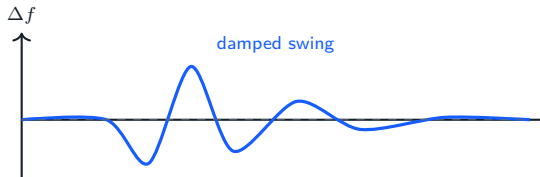


Used by final variant

The selected localizer includes `rls_kalman`; the non-selected `swing_ekf` branch was tested but not used in the final route.

Classical generator intuition

$$\dot{\delta} = \omega - \omega_0$$
$$M\dot{\omega} = P_m - P_e - D(\omega - \omega_0)$$



What the branch tried

Use a light extended-Kalman style dynamic residual to capture frequency and angle swings after generation/load disturbances.

Why it was not final

It improved some synthetic validation behavior but reduced RAW localization. The final route kept the simpler RLS/Kalman innovation block.

What is computed

- ▶ PMU spread max/mean/p95.
- ▶ Time-to-peak range and standard deviation.
- ▶ Pairwise PMU differences.
- ▶ Pairwise PMU correlations.
- ▶ Pairwise time-to-peak differences.
- ▶ Weighted graph differences using electrical distance.

Why it helps localization

The event origin leaves a spatial fingerprint: which PMU changed most, which changed first, and which pairs moved coherently.

Used most by

Line outages and bus localization, where spatial pattern matters more than one local amplitude.

For signal s and observed buses $\mathcal{B}$

$$\text{spread}_s(t) = \max_{b \in \mathcal{B}} \tilde{x}_{b,s}(t) - \min_{b \in \mathcal{B}} \tilde{x}_{b,s}(t)$$

$$\Delta_{ab,s}^{max} = \max_t |\tilde{x}_{a,s}(t) - \tilde{x}_{b,s}(t)|$$

$$\rho_{ab,s} = \text{corr}(\tilde{x}_{a,s}, \tilde{x}_{b,s})$$

$$\Delta t_{ab,s}^{peak} = |t_{a,s}^{peak} - t_{b,s}^{peak}|$$

Electrical weighting

$$w_{ab} = \frac{1}{\max(d_{ab}^{eff}, 10^{-5})}$$

$$G_s^{max} = \max_{a < b} w_{ab} \Delta_{ab,s}^{max}$$

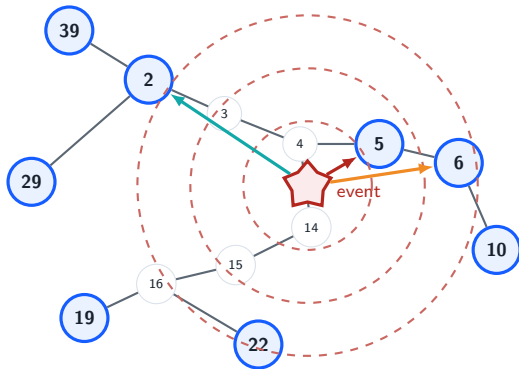
Feature names

GRAPH__VA_MAG__pmu_spread_max

GRAPH__Freq__pair_2_39__corr

GRAPH__IA_ANG__weighted_graph_diff_mean

Graph-temporal localization illustration



What a nearby PMU sees

Larger amplitude, earlier peak, and stronger correlation with electrically close PMUs.

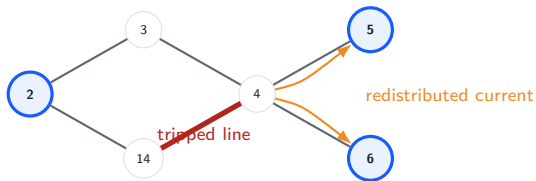
What a remote PMU sees

Smaller response, delayed peak, or lower pairwise coherence.

Line outage reading

For Event 2, the localizer learns which line candidates match the current redistribution and graph-difference pattern.

Why line outages need graph features



Electrical effect

Opening one line changes current paths. Voltage may not collapse like a fault, but current magnitude, current angle, and pairwise PMU differences move in a structured way.

Feature consequence

Event 2 benefits from graph differences, time-to-peak differences, and current-angle/current-magnitude signatures more than from a single largest voltage sag.

Selected

`base_v2 + rolling + rls_kalman + graph_ten`

$$n_{\text{features}} = 45,162$$

Not selected

- ▶ algebraic_residual_v4: matched RAW but did not improve it.
- ▶ swing_ekf: improved SIM slightly, reduced RAW localization.
- ▶ oracle topology ranker: better RAW, but not deployable.

Selection criterion

Highest RAW0001 localization among production-feasible variants while keeping strong synthetic validation performance.

Section 5

Hierarchical ML Method



Pipeline

$X \rightarrow \text{SimpleImputer}(\text{median}) \rightarrow \text{ExtraTreesClassifier}$

Imputation

Median imputation handles NaNs and missing feature values without requiring a complex preprocessing model.

Why ExtraTrees worked well

- ▶ Handles many features.
- ▶ Does not require scaling.
- ▶ Captures nonlinear interactions.
- ▶ Tolerates noisy/outlier-prone features.
- ▶ Fast enough for repeated experiments.

ExtraTrees decision rule

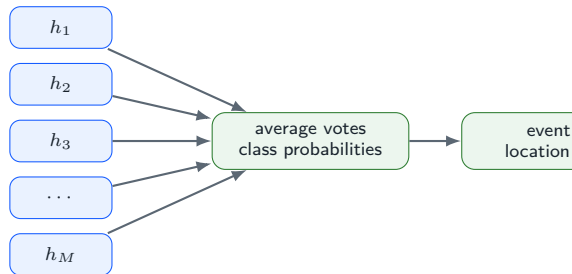
Ensemble probability

$$\hat{p}(c | x) = \frac{1}{M} \sum_{m=1}^M \mathbb{I}[h_m(x) = c]$$

$$\hat{c} = \arg \max_c \hat{p}(c | x)$$

What makes ExtraTrees different

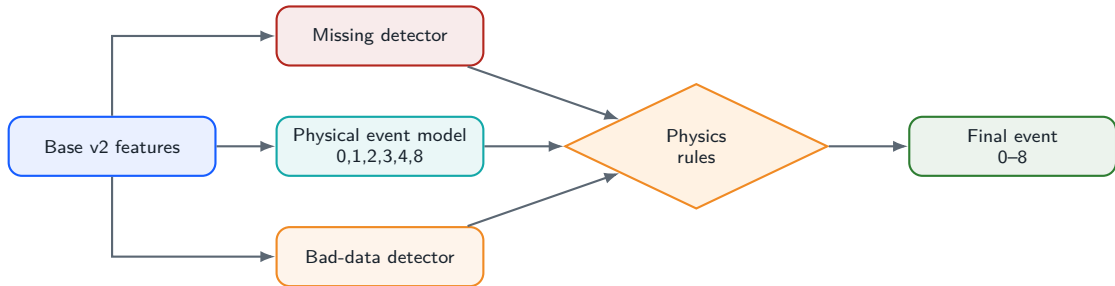
Each tree uses randomized split candidates. That reduces variance and works well when many engineered features are redundant but useful.



Pipeline detail

Before the forest, missing values are replaced by the training median of each feature column.

Model artifact	Role	Output
physical_event_classifier.joblib	Main physical event classifier	0,1,2,3,4,8
bad_data_detector.joblib	Event 7 detector	bad / not bad
missing_composition_detector.joblib	Event 5/6 support	missing composition
final_dynamic_localizers.joblib	Typed localizers	BUS, LINE, PMU
hierarchical_model_config.json	Rule thresholds	final event logic



Why hierarchical?

Missing data and bad data are measurement phenomena. They should not be forced into the same flat classifier decision as physical grid events.

Fault-like

$$V_{\text{span}} \geq 50,000$$
$$\max |dV/dt| \geq 500,000$$

Line-outage-like

$$I_{\text{span}} \geq 500$$
$$\max |dI/dt| \geq 3,000$$
$$V_{\text{span}} \leq 50,000$$

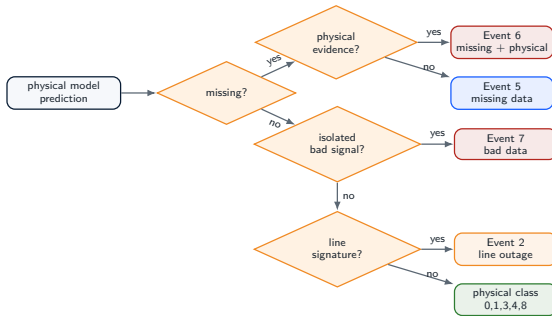
Missing-like

$\text{data_present_fraction} < 0.999$
or high NaN/gap features.

Bad-data-like

$$\text{single_signal_score} \geq 200,000$$
$$\text{single_signal_ratio} \geq 3.0$$

Rule engine as a decision tree



What this diagram leaves out

The implementation also uses confidence gates so weak physical predictions do not override strong missing or bad-data evidence.

1. Predict the physical event class using the physical ExtraTrees model.
2. Compute physical confidence from predicted probabilities.
3. Check missing-data evidence from data-quality features.
4. Check bad-data evidence from single-signal anomaly features.
5. Use fault and line-outage guards to correct ambiguous physical predictions.
6. If missing is present, decide Event 5 vs Event 6 using physical evidence.
7. If isolated bad-data evidence dominates, assign Event 7.

Interpretability benefit

The final decision can be defended physically: voltage fault, current redistribution, missing data, or isolated sensor corruption.

Section 6

Event-by-Event Interpretation



Main evidence

- ▶ Voltage magnitude span.
- ▶ Voltage derivative.
- ▶ Robust z-score in voltage channels.
- ▶ Early segment changes.
- ▶ Cross-PMU severity pattern.

Manual recipe

1. Build baseline from pre-event voltage.
2. Detect large voltage sag/transient.
3. Reject missing-only and bad-data-only cases.
4. Localize to BUS using spatial severity.

Main evidence

- ▶ Current magnitude span.
- ▶ Current derivative.
- ▶ Current-angle changes.
- ▶ Voltage span not as extreme as a fault.
- ▶ Graph-temporal pairwise PMU differences.

Manual recipe

1. Look for current redistribution.
2. Guard against fault-like voltage collapse.
3. Use graph features to score candidate lines.
4. Localize to LINE.

Main evidence

- ▶ Frequency shift.
- ▶ ROCOF response.
- ▶ Active-power proxy change.
- ▶ Late/pre sustained delta.
- ▶ RLS/Kalman residuals.

Physical explanation

Generation changes disturb the active-power balance. The grid response is often visible in frequency, ROCOF, and power proxy rather than only voltage magnitude.

Main evidence

- ▶ Current magnitude level shift.
- ▶ Active/reactive power proxy.
- ▶ Power-factor angle.
- ▶ Late segment persistence.
- ▶ Less abrupt than a fault.

Physical explanation

Load changes alter consumption. They often appear as sustained current and power-proxy shifts with less violent voltage transients.

Event 5: missing data

- ▶ DATA_PRESENT drops.
- ▶ Timestamp gaps grow.
- ▶ NaN fraction increases.
- ▶ No strong physical event evidence.

Event 6: missing + physical

- ▶ Missing-data evidence exists.
- ▶ Physical classifier is not normal.
- ▶ Physical confidence is sufficient.
- ▶ Location may be BUS or LINE depending on physical event.

Main evidence

- ▶ One signal has extreme robust z-score.
- ▶ Top anomaly is much larger than second anomaly.
- ▶ Other PMUs/channels remain coherent.
- ▶ Missing features do not dominate.

Localizer

Event 7 localizes to PMU, not BUS or LINE. The object that failed is the measurement device/channel source.

Main evidence

Event 8 captures cases with physical evidence that does not cleanly fit fault, line outage, generation change, or load change.

Useful features

- ▶ Power proxy.
- ▶ Rolling dynamics.
- ▶ RLS/Kalman residuals.
- ▶ Graph-temporal pattern.

Routing

The final runtime routes Event 8 to a BUS localizer because it represents a physical-grid signature rather than a PMU-only failure.

Section 7

Localization



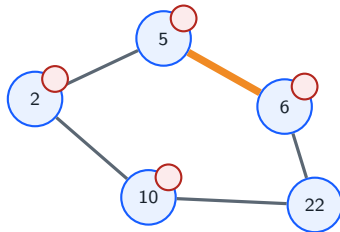
Event routing

Event	Localizer type
1 fault	BUS
2 line outage	LINE
3 generation change	BUS
4 load change	BUS
5 missing data	PMU
6 missing + physical	BUS or LINE
7 bad data	PMU
8 mixed proxy	BUS

Routing reason

A line outage, a bus fault, and a bad PMU are different localization problems. A single flat location classifier would mix incompatible targets.

Three location spaces



BUS target
fault, generation,
load, mixed

LINE target
line outage

PMU target
missing or bad data

same PMU measurements, different target sets

How to explain it

The location column does not always mean a bus. Its meaning depends on the event class.

Localizer keys in the final dynamic bundle

BUS, BUS:event1, BUS:event3, BUS:event4, BUS:event8,
LINE, LINE:event2, PMU

Model family

Each localizer is a median-imputed ExtraTrees pipeline trained on base + dynamic features.

Final Top-1

`use_hybrid_top1 = false`. The final submitted Top-1 location uses the validated typed ML localizers, not the hybrid/oracle branch.

BUS	LINE	PMU
▶ nearest severe response, ▶ voltage/frequency signature, ▶ graph spread pattern.	▶ current redistribution, ▶ pairwise PMU differences, ▶ topology-aware graph signature.	▶ missing run, ▶ isolated bad channel, ▶ local data-quality failure.

Location signal

Location is inferred from the pattern of who saw the disturbance, how strongly, and how early.

Localizer inference equation

Feature vector for location

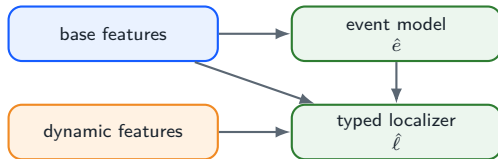
$$x_{\text{loc}} = [x_{\text{base_v2}}, x_{\text{rolling}}, x_{\text{rls/kalman}}, x_{\text{graph}}]$$

$$\hat{e} = H(x_{\text{base_v2}})$$

$$\hat{\ell} = L_{g(\hat{e})}(x_{\text{loc}})$$

Route function

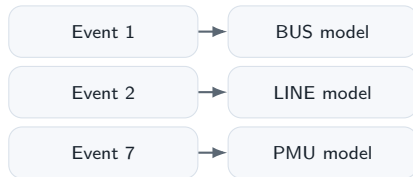
$$g(e) \in \{\text{BUS}, \text{LINE}, \text{PMU}\}$$



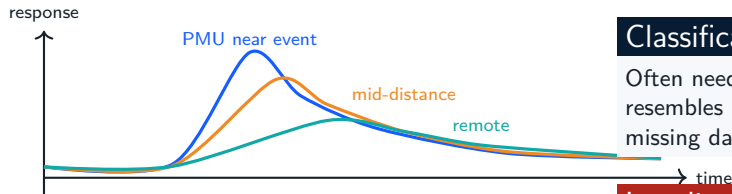
Reason for two feature views

The event classifier is stable with base features. The localizer gets the larger dynamic vector because location depends on propagation and topology.

BUS	LINE	PMU
$\mathcal{Y}_{BUS} = \{1, \dots, 39\}$ Target is the electrical bus where the physical disturbance is assigned.	$\mathcal{Y}_{LINE} = \{(i, j) : (i, j) \in \mathcal{E}\}$ Target is a branch in the IEEE 39 topology.	$\mathcal{Y}_{PMU} = \{2, 5, 6, 10, 19, 22, 29, 39\}$ Target is the observed device affected by missing or bad data.



What makes localization harder than classification



Classification

Often needs to know whether the pattern resembles a fault, outage, load change, missing data, or bad data.

Localization

Needs the finer spatial ordering: which PMU changed first, which changed most, and which candidate location best explains that pattern.

Section 8

Training, Validation, and Results



Training sources

- ▶ Synthetic SGSMA scenarios.
- ▶ Base features: `factory_features_v2.csv`.
- ▶ Dynamic features:
`dynamic_features_v3.csv`.
- ▶ Random state: 20260503.

Validation

- ▶ 30% stratified SIM holdout.
- ▶ RAW0001 transfer validation.
- ▶ Metrics: detector, classifier, macro-F1, localization.

Generalization principle

RAW labels were used for validation and analysis, not to train the final production-feasible submission path.

SIM holdout

Metric	Score
Detector accuracy	0.9693
Classifier accuracy	0.9673
Classifier macro-F1	0.9598
Localizer Top-1	0.8524

RAW0001

Metric	Score
Detector accuracy	1.0000
Classifier accuracy	1.0000
Classifier macro-F1	1.0000
Localizer Top-1	0.6667

Interpretation

Detection and classification transferred perfectly on RAW0001; localization remained the hard part.

Why the final variant was selected



Selected variant

`base_v2 + rolling + rls_kalman + graph_temporal`

Strength

- ▶ Strong SIM performance.
- ▶ Best RAW localization tier among deployable variants.
- ▶ No oracle RAW label use.
- ▶ Compact enough for submission.

Tradeoff

Some experimental methods had higher RAW numbers, but were not valid for unseen data because they used RAW true labels or oracle assumptions.

DeepSeek role

DeepSeek-style experiments were an exploration branch. They exposed RAW behavior and candidate localization ideas, but they were not the submitted inference route.

Why they were not the final method

The best RAW numbers from some DeepSeek variants used RAW labels directly for training, oversampling, weighting, or oracle ranking. That is not a valid inference setting for a new dataset.

Submitted method

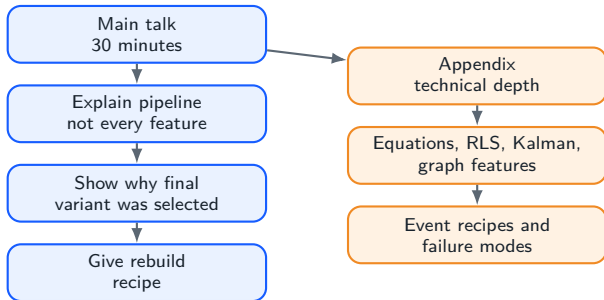
The final submitted method was SGMS: physics-informed features + hierarchical ExtraTrees + typed dynamic localizers.

Section 9

Presentation Script and Coding Recipes



Use the deck as two decks

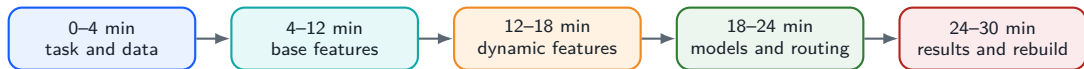


Presentation rule

The live talk should show the method as a buildable system. The appendix should answer detailed implementation questions.

Practical plan

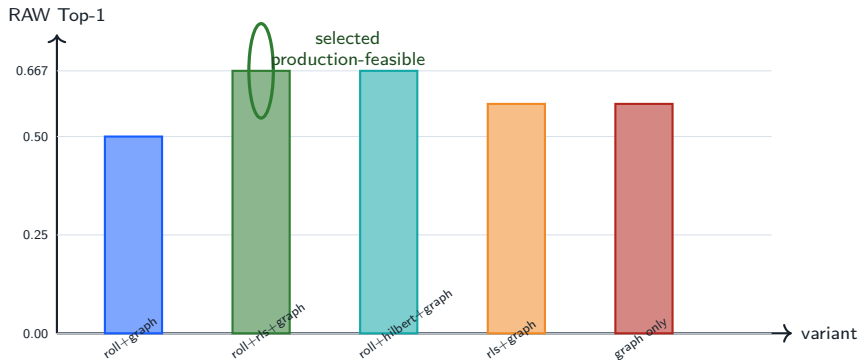
Use about 35 slides live. Keep the rest ready for questions about math, feature names, event routing, or DeepSeek.



One sentence to repeat

We do not classify raw waveforms directly. We turn PMU windows into physics-aware features, classify the event hierarchically, then localize it in the right target space.

Ablation chart: why the selected variant survived



Reading

The selected model kept perfect RAW detection/classification and reached the best deployable RAW localization tier without using RAW labels as training signal.

Equation to code to feature: robust abnormality



Equation

$$m = \text{median}(x_{pre})$$
$$s = 1.4826 \text{ median}|x_{pre} - m|$$
$$z(t) = \frac{|x(t) - m|}{\max(s, \epsilon)}$$

Code shape

```
m = median(pre)
s = 1.4826 * mad(pre)
z = abs((x-m)/max(s, eps))
```

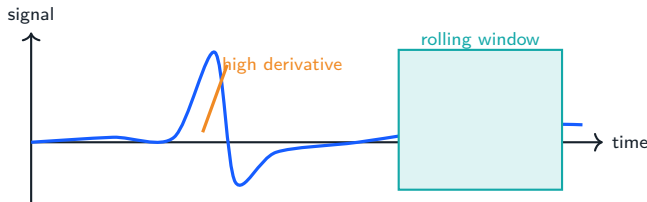
Feature names

```
max_abs_robust_z
sustained_abs_robust_z_5
sustained_abs_robust_z_15
```



Derivative

$$d_x(t_i) = \frac{\tilde{x}(t_i) - \tilde{x}(t_{i-1})}{\Delta t}$$
$$\max |d_x|, \quad \sqrt{\frac{1}{T} \sum_i d_x(t_i)^2}$$



Feature names

max_abs_derivative
rms_derivative
ROLL__w15__slope_max_abs

Code shape

```
dx = gradient(centered, dt)
roll = rolling(centered, w).mean()
roll_slope = gradient(roll, dt)
```

RLS residual

$$\hat{y}_t = \phi_t^\top \theta_{t-1}, \quad r_t = |y_t - \hat{y}_t|$$
$$\theta_t = \theta_{t-1} + K_t(y_t - \hat{y}_t)$$

Feature names

RLSK_*level_resid__max
RLSK_*linear_resid__p95
RLSK_*theta_delta__late_pre_delta

Kalman innovation

$$\nu_t = y_t - \hat{x}_{t-1}$$
$$K_t = \frac{P_t^-}{P_t^- + R}, \quad \hat{x}_t = \hat{x}_{t-1} + K_t \nu_t$$

Feature names

RLSK_*kalman_innov__max
RLSK_*kalman_innov__mean
RLSK_*kalman_innov__time_to_peak



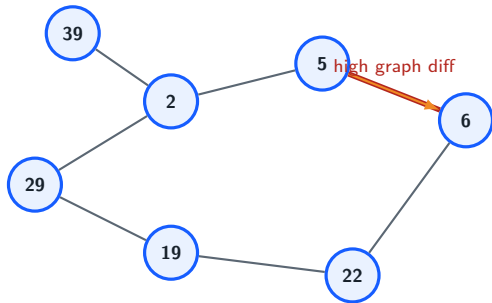
Severity and pair relation

$$S_b = \max_s \max_t |z_{b,s}(t)|$$

$$\Delta_{ab,s} = \max_t |\tilde{x}_{a,s}(t) - \tilde{x}_{b,s}(t)|$$

$$w_{ab} = \frac{1}{\max(d_{ab}^{eff}, 10^{-5})}$$

$$G_{ab,s} = w_{ab} \Delta_{ab,s}$$



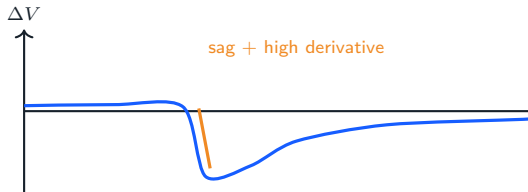
Feature names

GRAPH_*pmu_spread_max

GRAPH_*pair_a_b_corr

GRAPH_*weighted_graph_diff_mean

Event 1 recipe: fault



Minimum detector

$$\max |z_V| \uparrow, \quad \max \left| \frac{dV}{dt} \right| \uparrow$$

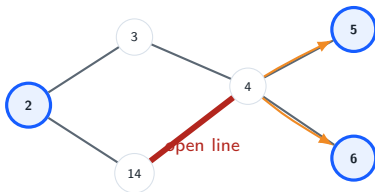
Use these features

VA/VB/VC_MAG__early__span
max_abs_derivative
sustained_abs_robust_z_5
GRAPH__*time_to_peak_*

Location

Route to BUS localizer. The bus target is inferred from severe nearby PMU response and graph timing.

Event 2 recipe: line outage



Minimum detector

Look for current redistribution and angle/current changes without the fault-level voltage sag.

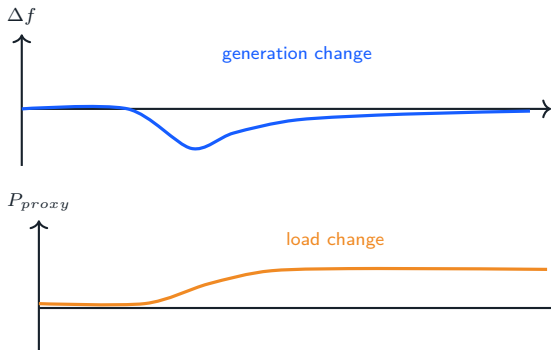
Use these features

IA/IB/IC_MAG__*
IA/IB/IC_ANG__*
P/Q_PROXY__*
GRAPH__*weighted_graph_diff_*

Location

Route to LINE localizer. The target is a branch (i, j) , not a bus.

Events 3 and 4 recipes: generation and load



Event 3

Generation change: frequency/ROCOF and power proxy shift. Route to BUS localizer.

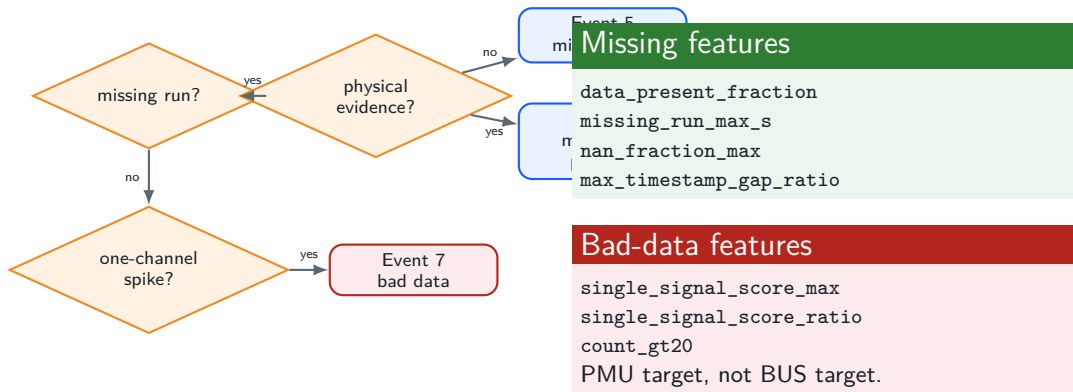
Event 4

Load change: sustained current and power proxy level shift. Route to BUS localizer.

Use these features

```
Freq__late__mean  
ROCOF__early__span  
P_PROXY__*late_pre_delta  
RLSK__*linear_resid_*
```

Events 5, 6, and 7 recipes: data failures



Event 8 recipe: mixed physical proxy

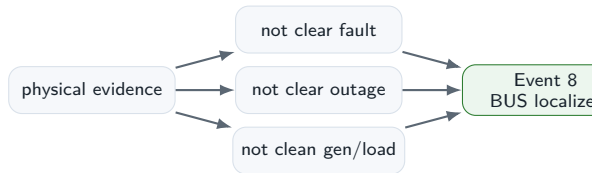


When it appears

The window has physical evidence, but it does not cleanly match fault, line outage, generation change, or load change.

Feature signature

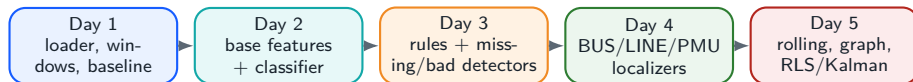
power proxy shifts
frequency/ROCOF evidence
graph spread
RLS/Kalman innovation



Location

Route to BUS. It is treated as a physical-grid event, not a PMU-only failure.

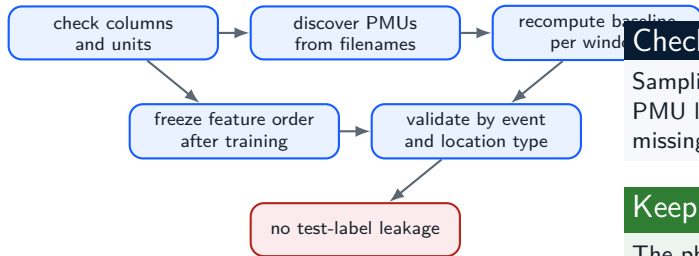
Five-day rebuild plan for a new dataset



Competition rule

Build a working baseline early. Add dynamic features only after the event classifier and typed localizers produce valid CSV output.

Adapting to a new RAW dataset



Check first

Sampling rate, timestamp continuity, PMU list, channel names, angle units, and missing-data coding.

Keep stable

The physical feature definitions should stay the same. Retrain models and recalibrate thresholds if the new data distribution shifts.

Section 10

Hands-on Reimplementation



```
case = read_pmu_case(input_dir)
windows = split_into_windows(case, seconds=30)

for window in windows:
    x_base = extract_window_features_v2(window)
    x_dyn = extract_dynamic_features(
        window,
        include_blocks=("rolling", "rls_kalman", "graph_temporal"),
    )

    physical = physical_model.predict(x_base)
    bad_score = bad_data_model.predict_proba(x_base)
    missing_score = missing_model.predict_proba(x_base)

    event = apply_physics_rule_engine(
        physical, bad_score, missing_score, x_base
    )

    loc_type = route_event_to_localizer(event, physical)
    location = localizers[loc_type].predict(concat(x_base, x_dyn))

    emit_rows_for_all_timestamps(window, event, location)
```

Build order if you must code it from scratch



1. Load all `Bus*.csv` files and align timestamps.
2. Create fixed 30-second windows.
3. Compute pre-event medians and unwrap angles.
4. Extract segment statistics, temporal deltas, and derivatives.
5. Add robust z-score and sustained anomaly features.
6. Add data-quality and single-signal bad-data features.
7. Add power proxy features.
8. Train physical ExtraTrees classifier.
9. Train missing and bad-data detectors.
10. Add physics rule guards.
11. Train BUS, LINE, and PMU localizers.
12. Add rolling and graph-temporal features.
13. Add RLS/Kalman if time remains.

Implement first

- ▶ 30 s windows.
- ▶ Pre-event baseline.
- ▶ Segment statistics.
- ▶ Derivatives.
- ▶ Robust z-score.
- ▶ Data-quality features.
- ▶ Power proxy.
- ▶ ExtraTrees + rules.

Then add

- ▶ Rolling dynamics.
- ▶ Graph-temporal PMU comparisons.
- ▶ RLS/Kalman residuals.
- ▶ Event-specific localizers.
- ▶ Top-K diagnostics.

Practical rule

Detection/classification can be strong with base features. Localization benefits the most from dynamic and graph-temporal features.

What to memorize for the defense



1. The decision unit is a 30-second PMU window.
2. The baseline is the pre-event median.
3. The features are physics-aware: V, I, angle, frequency, ROCOF, quality, and power proxy.
4. Dynamic features add temporal evolution and spatial propagation.
5. The classifier is hierarchical, not flat.
6. ExtraTrees is used because it handles many nonlinear, noisy features.
7. Localization is typed: BUS, LINE, or PMU depending on event.
8. DeepSeek was exploratory; SGMS was the final valid submitted method.

Final method in one sentence

The final submission transforms raw PMU windows into physically meaningful feature vectors, then uses hierarchical ExtraTrees models and typed dynamic localizers to produce event and location predictions.

Feature engine

Base v2 + rolling +
RLS/Kalman + graph temporal.

Classifier

Physical event model + missing
detector + bad-data detector +
rules.

Localizer

Specialized BUS, LINE, and
PMU ExtraTrees localizers.

Final variant: `base_v2 + rolling + rls_kalman + graph_temporal`

Backup

Detailed Reference



Runtime bundle

- ▶ `models/final_model_config.json`
- ▶ `models/feature_columns.json`
- ▶ `models/classifiers/physical_event_classifier.joblib`
- ▶ `models/detector/bad_data_detector.joblib`
- ▶ `models/detector/missing_composition_detector.joblib`
- ▶ `models/detector/hierarchical_model_config.json`
- ▶ `models/localizer/final_dynamic_localizers.joblib`

Submission note

The large redundant hybrid localizer artifact was excluded from the compact zip; the final dynamic bundle is enough for the submitted Top-1 path.

Parameter	Value
bad data probability	0.4211785821
missing composition probability	0.0
missing physical confidence	0.0
fault voltage span min	50,000
fault voltage derivative min	500,000
line current span min	500
line current derivative min	3,000
line voltage span max	50,000
bad single-signal score min	200,000
bad single-signal ratio min	3.0
line gate max physical confidence	0.75
fault guard max physical confidence	0.60

Final main classifiers

- ▶ Physical event classifier: ExtraTrees, 750 estimators, `max_features="sqrt"`, balanced class weights.
- ▶ Bad-data detector: ExtraTrees, 650 estimators, `max_features="sqrt"`, balanced class weights.
- ▶ Missing-composition detector: ExtraTrees, 650 estimators, `max_features="sqrt"`, balanced class weights.
- ▶ Dynamic localizers: ExtraTrees pipelines, 180 estimators in the final dynamic bundle.

Common preprocessing

`SimpleImputer(strategy="median")` before each ExtraTrees model.

Generic inference

```
python main.py -input-dir <RAW_OR_SIM_FOLDER> -output <OUTPUT_CSV>
```

SIM smoke test

```
python main.py -input-dir workbench\simulated\sgsma_generated\SIM00642\pmu -output  
workbench\_tmp\SIM00642_predictions.csv
```

RAW0001

```
python main.py -input-dir data\RAW0001 -output data\RAW0001\predictions.csv
```

Test suite

```
python -m pytest    Last reported local run: 246 passed, 10 skipped
```

Backup: how to answer a generalization question



Answer

The method is designed to generalize because it uses physical feature transformations, simulated training scenarios, hierarchical rules, and RAW validation only as evaluation evidence.

Main risk

Localization is more sensitive than detection/classification because the spatial fingerprint depends on PMU placement, topology, and event location.

What to improve on a new dataset

Calibrate localizers and topology-aware guardrails using valid training labels from the new dataset, while avoiding test-label leakage.